



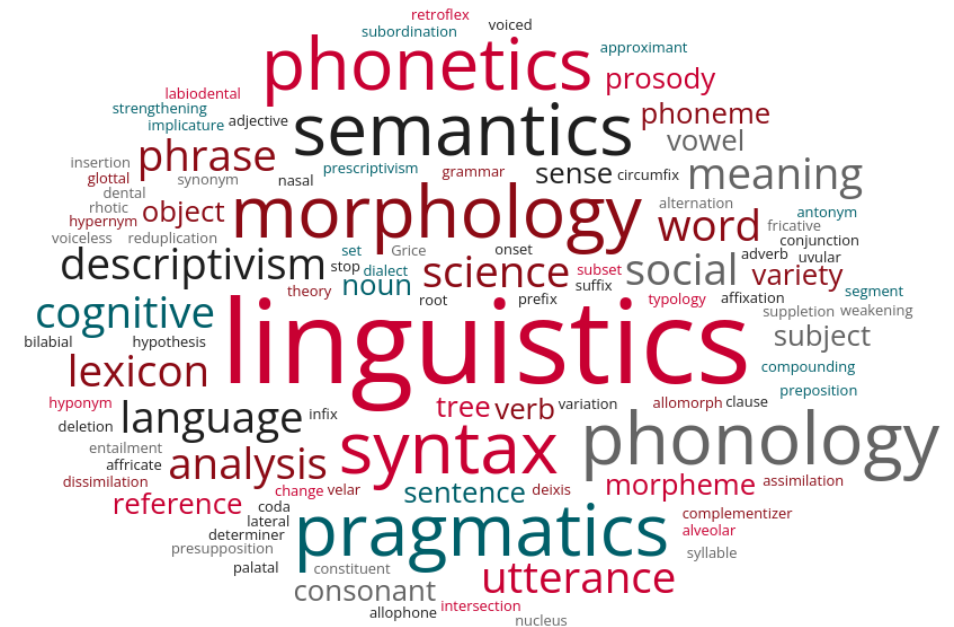
# Morphology I

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2026/09/09 (updated: 2026-09-09)

Please pick up your **name card** from the front

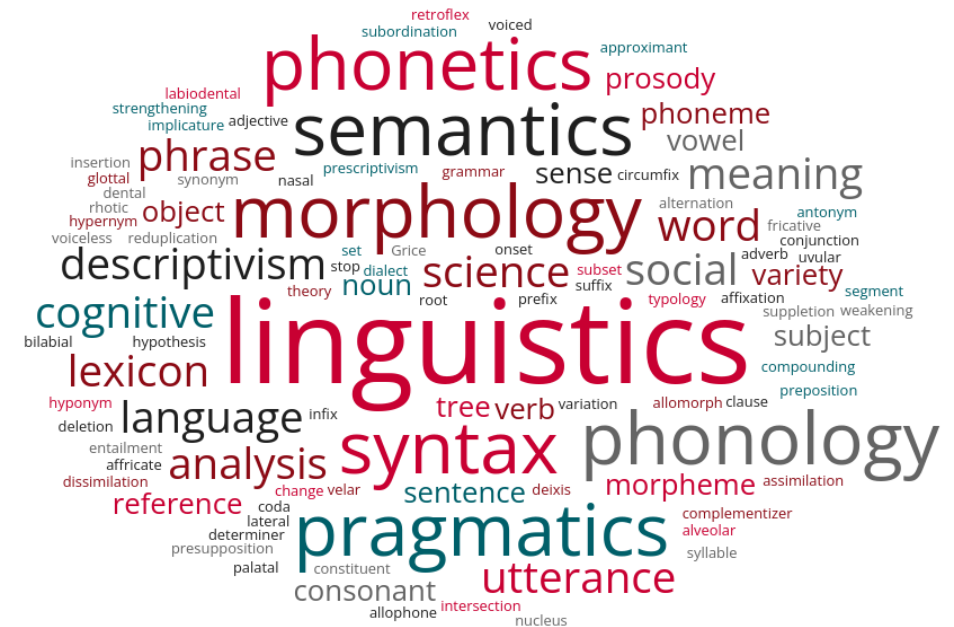
# Previously on 201: What is Linguistics

- Linguistics is the \_\_\_\_\_ study of language
- linguists use the scientific method to uncover the \_\_\_\_\_ or knowledge of language users
- while there are both prescriptive and descriptive rules relating to language, linguists are interested in uncovering the \_\_\_\_\_ rules of a language
- often times these rules are \_\_\_\_\_, hence speakers are not aware that they are following them
- we can use the scientific method to uncover rules of languages we ourselves speak or languages we are unfamiliar with

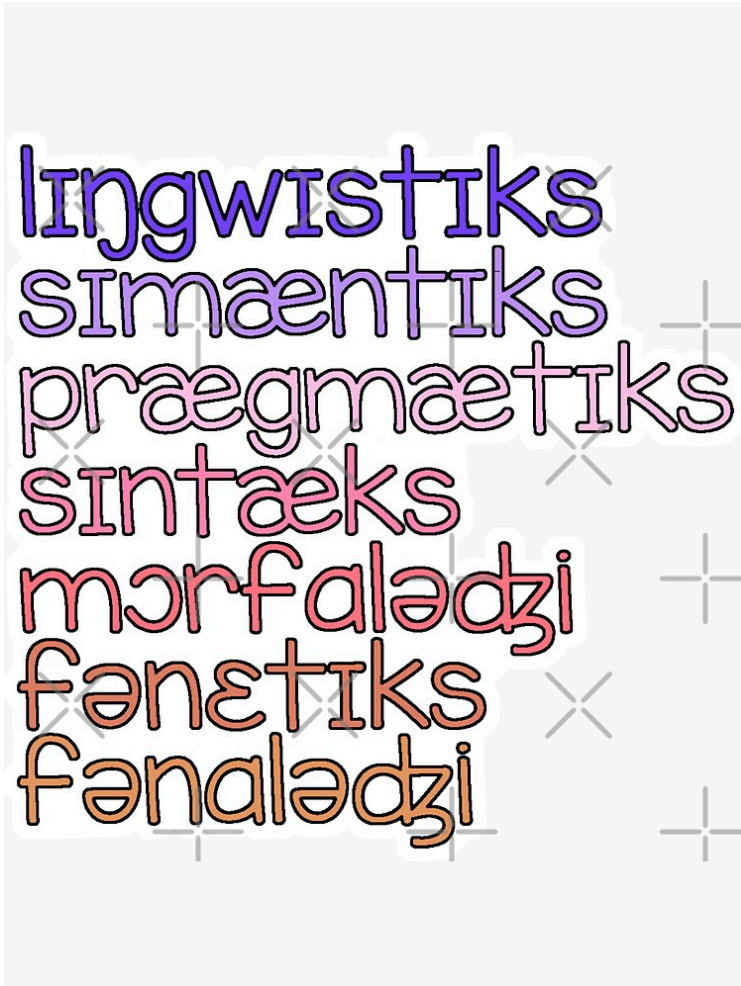


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# Different Parts of Linguistic Competence



- Linguistics is the scientific study of human languages
  - Morphology studies word structure
  - **Syntax** studies **sentence** structure
  - **Semantics** studies **literal meaning** of words and sentences
  - **Pragmatics** studies **contextual meaning** of words and sentences
  - **Phonetics** studies speech **sounds**
  - **Phonology** studies **sound patterns**

# Morphology

# What are Words

as we said, Morphology studies **word structure**, but what are words?

## word 1 of 2 noun

'wɜrd ⓘ

[Synonyms of word >](#)

- 1 a (1)** : a speech sound or series of speech sounds that symbolizes and communicates a meaning usually without being divisible into smaller units capable of independent use
- | the order of *words* in a phrase
  - | the meaning of a *word*
  - | a *word* that rhymes with her name
- (2)** : a gesture or series of gestures that symbolizes and communicates a meaning of an object or concept : **SIGN sense 1c**
- | ... signage shows visitors how to sign the *word* "goat"—while a goat pokes his head through the fence.
  - Reece Barrett
- (3)** : the entire set of linguistic forms produced by combining a single base with various inflectional elements without change in the part of speech elements
- | The *word* *go* also has the forms *goes*, *going*, *went*, and *gone*.

- 1 b (1)** : a written or printed character or combination of characters representing a spoken word
- | the number of *words* to a line
  - sometimes used with the first letter of a real or pretended taboo word prefixed as an often humorous euphemism
  - | ... the first man to utter the *f word* on British TV ...
  - Time*
  - | we were not afraid to use the *d word* and talk about death
  - Erma Bombeck
- (2)** : any segment of written or printed discourse ordinarily appearing between spaces or between a space and a punctuation mark

definition of '**word**' from *Merriam Webster Dictionary*

# What are Words

dog

[dag] (Amer.) / [dɒg] (Brit.)



# Some Relations are Arbitrary

- as pointed out by **Ferdinand de Saussure** the relationship between sound/sign and meaning appears to be **arbitrary**
- the fact that the sound [dag] has a meaning ‘a domesticated carnivorous mammal’ does NOT appear to rely on special properties of those sounds in that order
- what is ‘a domesticated carnivorous mammal’ in other languages?



*Ferdinand Mongin de Saussure (1857-1913)*

# Looking at Different Languages

- sound-meaning arbitrariness becomes clear when we look at **other languages**
- the **same meaning** associated with **different sounds**
  - inu (Japanese)
  - gae (Korean)
  - aso (Tagalog)
  - hundur (Icelandic)
  - Hund (German)...
- or the **same sound** associated with **different meaning**
  - [q<sup>h</sup>urt] in **Kazakh**  
‘insect’

◦ type of food made from milk’



*Inuyasha*

# How are words stored in our mind?

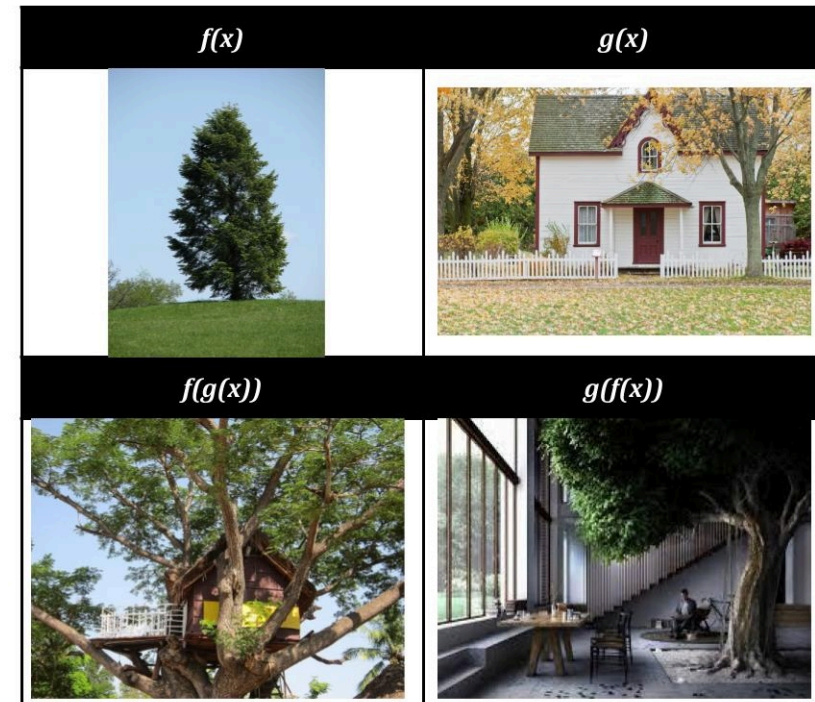
# Mental Dictionary (Lexicon)

- part of language learning then is learning and memorising these **meaning-sound pairs** and storing them in your **mental dictionary** (aka **lexicon**)
- each item in the lexicon is a **lexical entry**
- we know that part of each lexical entry is its *phonological representation* (written in IPA) and its *semantic representation*

| label                              | 'dog'           | 'teach'               |
|------------------------------------|-----------------|-----------------------|
| <b>phonological representation</b> | /dag/           | /tɪ:tʃ/               |
| <b>semantic representation</b>     | {x: x is a dog} | {<x, y>: x teaches y} |

# Compositionality

- **NOT** all pairs of sounds and meanings are arbitrary (actually most aren't)
- the following string of phones (sounds) has a **predictable** meaning
- and is almost certainly **NOT** stored as a single lexical entry
  - [mɜ:lɪns tɪtʃər wɒkt kwɪkli]
  - 'Merlin's teacher walked quickly.'
- the meaning of this string is predictable from the meanings of its parts
- it is **COMPOSITIONAL**

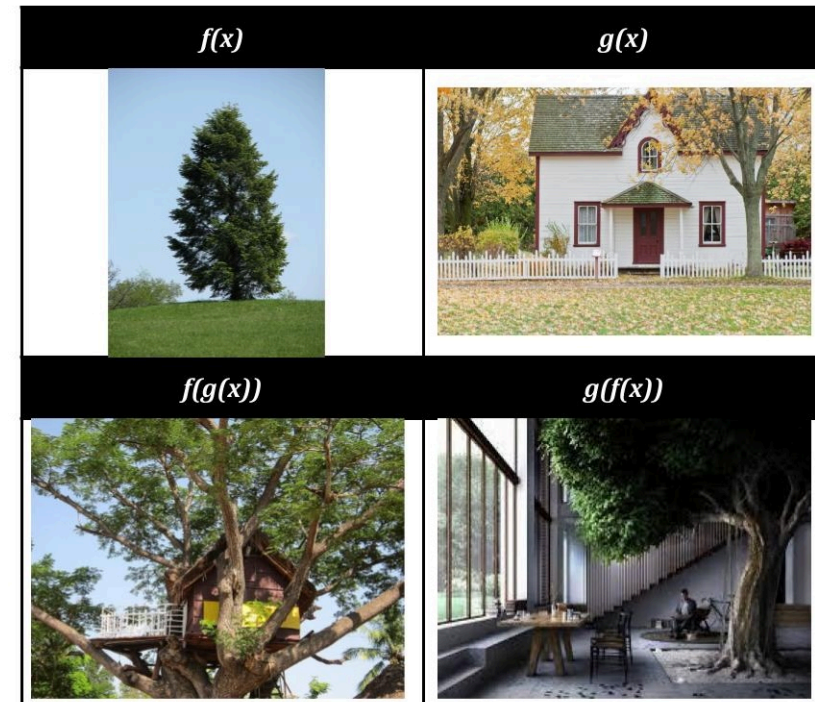


'compositionality'

# Compositionality

- [mɜ:lɪns tɪtʃər wɔkt kwɪkli]  
'Merlin's teacher walked quickly.'

How is each word in the sentence **compositional**?



'compositionality'

# Compositionality

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How is each word in the sentence **compositional**?

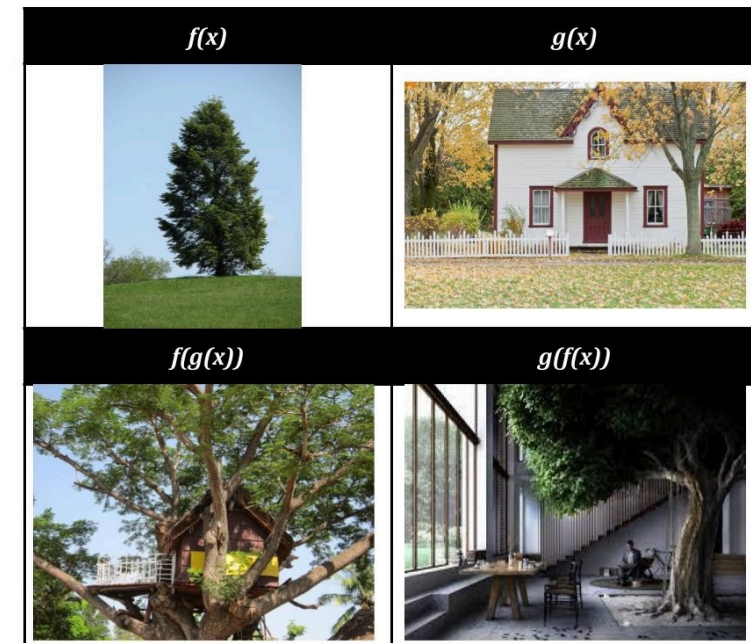
- words like **Merlin's** and **teacher CAN** be broken down
- and their meaning seems to contain the meaning of the pieces
- while a word like **dog, CANNOT** be broken down into smaller meaningful units
- it could be broken down into smaller sounds, but those sounds do not correspond to a meaning on their own

**Merlin's** and **teacher**

- morphologically complex words

**dog** and **teach**

- morphologically simple words



# How many meaningful units?

teacher

presidential

unlockable

antidisestablishmentarianism

# How many meaningful units?

teacher **teach-er**

presidential **president-ial**

unlockable **un-lock-able**

antidisestablishmentarianism  
**anti-dis-establish-ment-ar(y)-ian-ism**

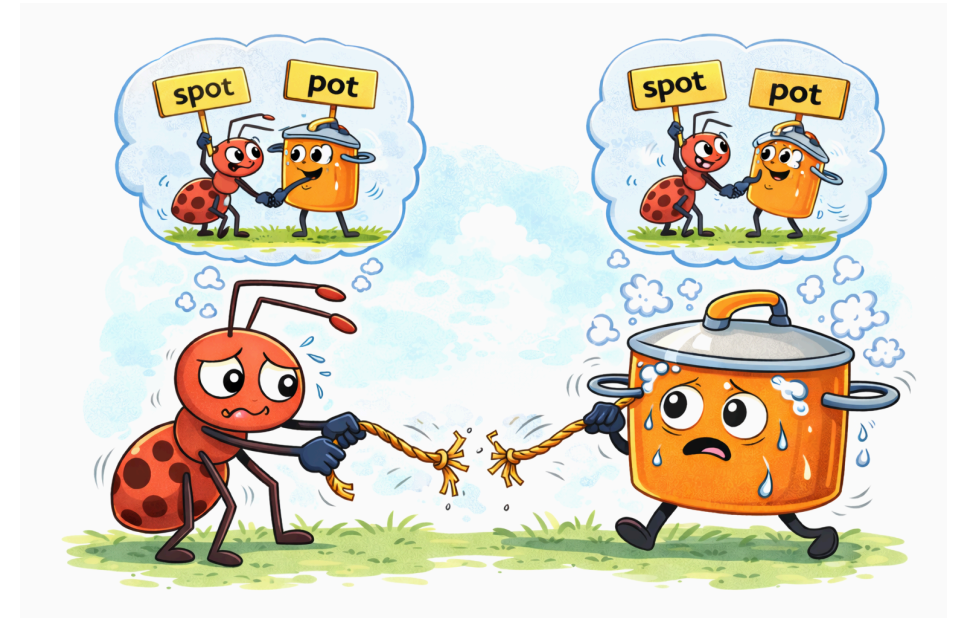
so we see systematic **PATTERNS**  
in the meanings of complex words

hospital, hospital**ise**, hospital**isation**

refute, refut**able**, **ir**refut**able**, irrefut**ability**

# Trickier Cases

- is the word **spot** complex because it contains the same sounds as the word **pot**? Why or why not?
- is the word **chef** complex because it appears to contain the meaning of the word **cook**? The pair **cook-chef** appears to be the same relation that holds between **teach-teacher**



# Zooming in on -er

- what do these words all have **in common** in terms of their meaning?
  - teach**er**
  - driv**er**
  - lectur**er**
  - read**er**
  - soften**er**
  - whiten**er**
  - blend**er**
  - modernis**er**
- -er seems to have a **shared and stable meaning**, something like '*one who does X action*', where X is a **verb**



# Zooming in on *-er*

- *-er* is used **productively**
- it can be used to create novel words
- the verb **yeet** is a fairly new verb
- if you know the meaning of **yeet**
- the term **yeeter** is easily understandable
- even if you have never heard it before

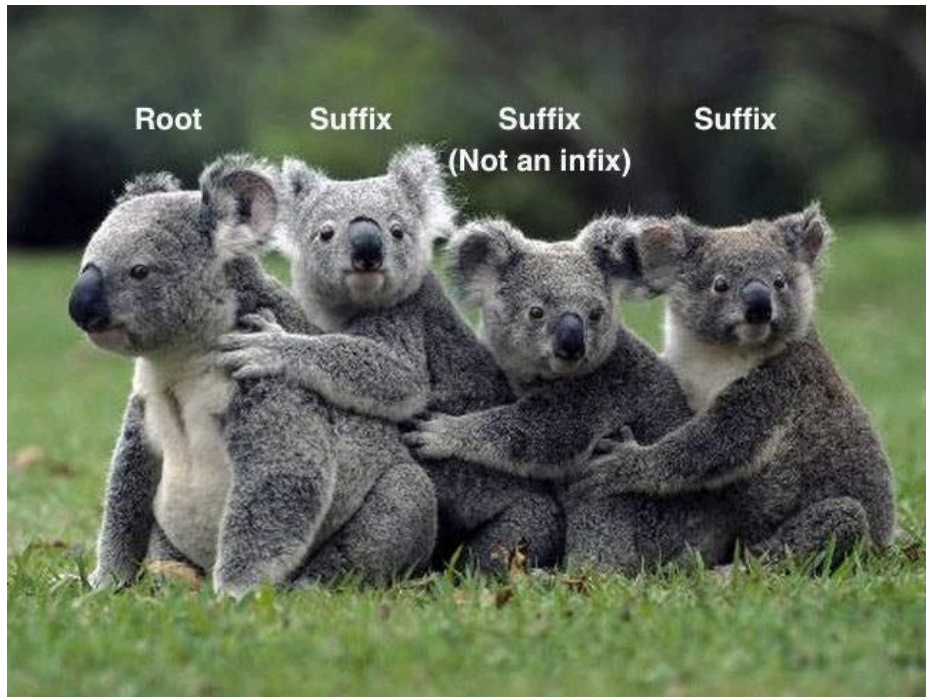


# Lexical Entries

| label                                  | '-er'                 | 'un-'        | '-able'                 |
|----------------------------------------|-----------------------|--------------|-------------------------|
| <b>phonological<br/>representation</b> | /əɹ/                  | /ʌn/         | /ʌbəl/                  |
| <b>semantic<br/>representation</b>     | one who does <i>X</i> | not <i>X</i> | able to be <i>X</i> -ed |

# Morphemes

teach, dog, fish and -er, un-, -able are all **morphemes**



## **morpheme**

- the minimal unit of phonology-meaning correspondence.
- in other words, a morpheme is a string of sounds whose meaning cannot be broken down into smaller meaningful units

# ***morpheme***

**smallest linguistic unit** with **meaning**

# Smallest Linguistic Unit with Meaning

- **smallest linguistic unit** means shortest sequence of sounds
- what's the **shortest** sequence of sounds with meaning you can **break** these words into?

*schoolbus*

*bee*

*elephant*

*opener*

*silliest*

# Smallest Linguistic Unit with Meaning

- we use **hyphens** (-) to indicate morpheme boundaries
- morphemes are **NOT** the same as syllables:
  - *elephant* has three syllables, but only **one** morpheme
  - *e*, *le*, *phant* don't mean anything – they're syllables, but not morphemes
- morphemes are made of **sounds**, NOT letters:
  - *silly* and *silli-* are two ways of spelling the same morpheme
  - same with *big*, *bigg*-er and *lie*, *ly*-ing
  - don't worry about spelling differences

*school-bus*

*bee*

*elephant*

*open-er*

*silli-est*

# Smallest Linguistic Unit with Meaning

- note that it's OK for a word to have **just one** morpheme
  - every word has at least one morpheme, some have more
- words with one morpheme are called **monomorphemic** or **simplex words**
  - *bee*, *elephant*
- words with two or more morphemes are **polymorphemic** or **complex words**
  - *school-bus*, *open-er*, *silli-est*
- sometimes you'll hear terms like **bimorphemic** or **trimorphemic**
  - what do you think they mean?

*school-bus*

*bee*

*elephant*

*open-er*

*silli-est*

# Smallest Linguistic Unit with **Meaning**

- morphemes can have two types of **meaning**:

## lexical meaning

- This is like a *dictionary definition*. (*lexicon* = mental dictionary)
- Consider these words: *apple-sauce*, *white-water*, *over-think*
- Morphemes like *apple*, *water*, *think* are easy to define: they have **lexical meaning**.

## grammatical function

- Now consider these words: *cat-s*, *laugh-ed*, *happi-ness*, *calm-ly*
- What do the morphemes *-s*, *-ed*, *-ness*, *-ly* mean?
- Rather than a dictionary definition, they have a **grammatical function**.
- Their meaning is related to *how the word behaves in a sentence*:  
*-s* = PLURAL, *-ed* = PAST, *-ness* = NOUN, *-ly* = ADVERB

# Exercise: Finding Morphemes 1

which of the following words contain the morpheme **un-** with the phonological form /ʌn/? And which are not?

unfaithful

unthinkable

untied

underwear

unlawful

unhappy

until

undershirt

untreated

unable

unsuspecting

unprepared

untouched

unclear

unpretentious

underprepared

# Exercise: Finding Morphemes 1

which of the following words contain the morpheme **un-** with the phonological form /ʌn/? And which are not?

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unsuspecting

unprepared

untouched

unclear

unpretentious

underprepared ✗

# Exercise: Finding Morphemes 2

break these words down into **morphemes**, then discuss with a partner:

- is *-lessness* a morpheme?
  - is *un-* a morpheme in both *untied* and *uncle*?
  - is *-s* a morpheme in both *sleeps* and *lens*?
  - does *er-* mean the same thing in both *smaller* and *helper*?
1. *sleeps*
  2. *sleepless*
  3. *sleeplessness*
  4. *lens*
  5. *smaller*
  6. *helper*
  7. *untied*
  8. *uncle*

# Exercise: Finding Morphemes 2

break these words down into **morphemes**, then discuss with a partner:

- is *-lessness* a morpheme?
  - **no**, it's two morphemes: *-less* and *-ness*
- is *un-* a morpheme in both *untied* and *uncle*?
  - **no**, only in *un-tie-d* because it means something there
- is *-s* a morpheme in both *sleeps* and *lens*?
  - **no**, only in *sleep-s* because it means something there
- does *er-* mean the same thing in both *smaller* and *helper*?
  - **no**, these morphemes are **homophones**

1. *sleeps*
2. *sleepless*
3. *sleeplessness*
4. *lens*
5. *smaller*
6. *helper*
7. *untied*
8. *uncle*

## Should complex words get their own lexical entry?

we have seen that both *teach* and *-er* have lexical entries  
should *teacher* also have one? why or why not?

|                             |                 |
|-----------------------------|-----------------|
| label                       | 'teacher'       |
| phonological representation | /tɪjtʃər/       |
| semantic representation     | one who teaches |



# Finding morphemes in other languages

## (Persian, Iranian)

try to match the meanings in (a) through (h) with a **morpheme** in the Persian data.

- a prefix is marked with a hyphen **following** the morpheme (X-)
- a suffix is marked with a hyphen **preceding** the morpheme (-X)

|               |                   |                    |                        |
|---------------|-------------------|--------------------|------------------------|
| (a) [xarid]   | 'bought'          | (e) [mixaridid]    | 'You (pl) were buying' |
| (b) [mixarid] | 'was buying'      | (f) [naxaridam]    | 'I did not buy'        |
| (c) [xaridam] | 'I bought'        | (g) [namixaridand] | 'They were not buying' |
| (d) [xaridi]  | 'you (sg) bought' | (h) [naxaridim]    | 'We did not buy'       |

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- a. I
- b. you (sg)
- c. we
- d. you (pl)

- e. they
- f. not
- g. PROG
- h. bought

# Finding morphemes in other languages

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|             |     |
|-------------|-----|
| a. I        | -am |
| b. you (sg) | i   |
| c. we       | im  |
| d. you (pl) | id  |

|           |       |
|-----------|-------|
| e. they   | -and  |
| f. not    | na-   |
| g. PROG   | mi-   |
| h. bought | xarid |

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| f. not    | na-   |
| g. PROG   | mi-   |
| h. bought | xarid |

is it possible to split **buy** and **PAST** from these data?

# Finding morphemes in other languages

## (Persian, Iranian)

|               |                   |                    |                        |
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How would you say the following in Persian?

- i. They were buying.
- ii. You (sg) did not buy.
- iii. You (sg) were buying.

# Finding morphemes in other languages

## (Persian, Iranian)

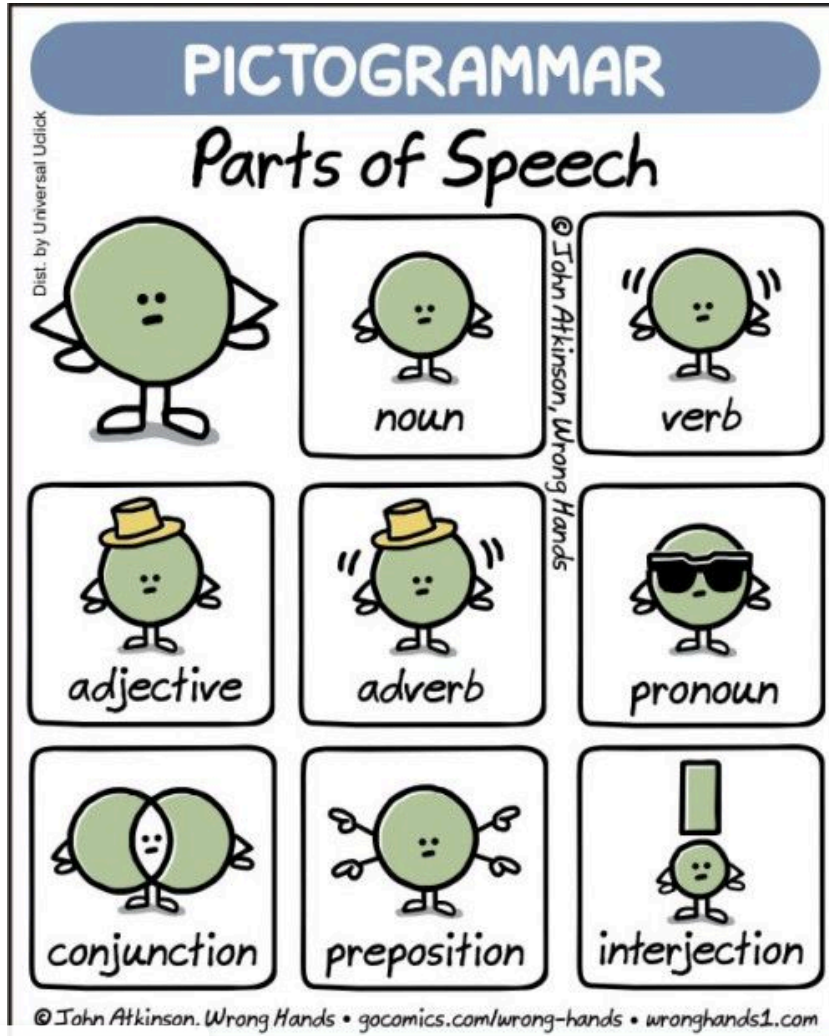
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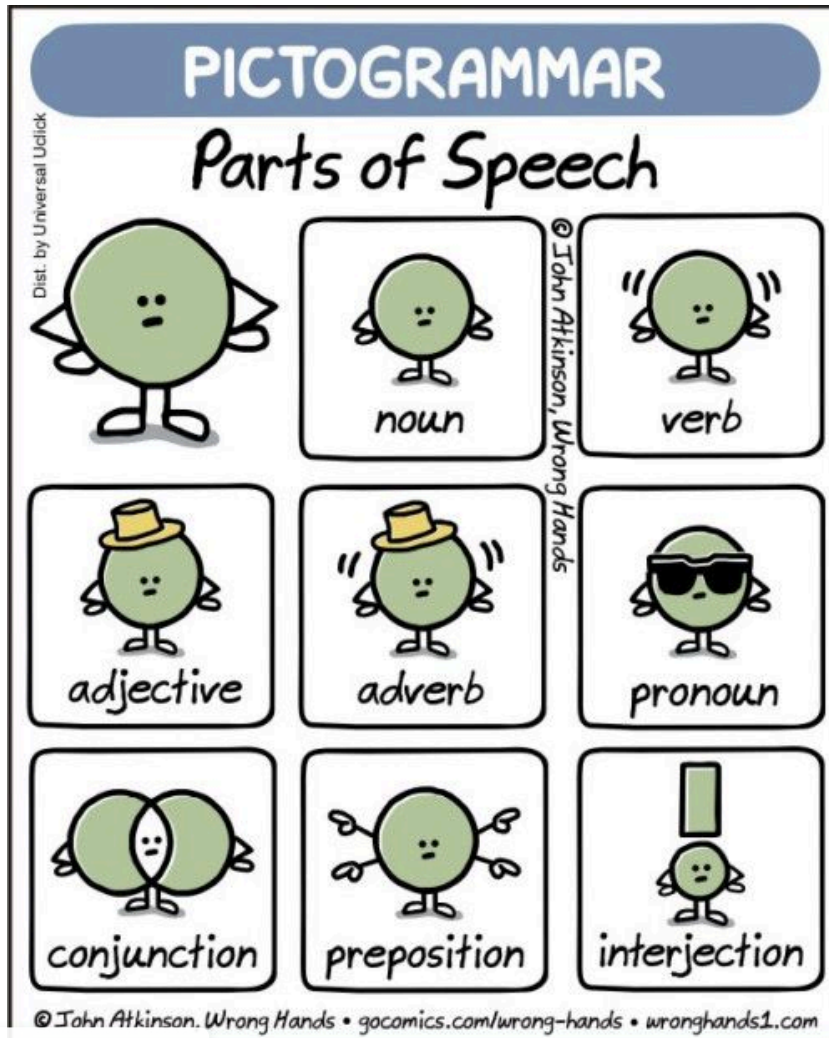
mi-xarid-and  
na-xarid-i  
mi-xarid-i

# Lexical Category / Parts of Speech



- **Noun (N.):** to name things, people, places, concepts, etc.
  - *water, sister, sky*
  - combine with plural morpheme -s;
  - often ends with -ion/-ment/...;
  - occur after a determiner, ...
- **Verb (V.):** to show an action or a state
  - *play, think, hit*
  - combine with past morpheme -d, has combined with -ise, ...
  - follow an auxiliary (does), modal (might), or negation (not), ...

# Lexical Category / Parts of Speech



- **Adjective (Adj.):** to describe a noun, a pronoun or part of a sentence
  - *tall, thin, smart*
  - comparative e.g., bigger
  - superlative e.g., biggest
  - combine with un-, has combined with -able, ...
  - occur between a determiner and a noun, ...
- **Adverb (Adv.):**
  - *carefully, intentionally, happily*
  - appear with -ly, ...
  - can't appear between a determiner and a noun, ...

# Lexical Category / Parts of Speech

How do we know that **wabe** is a **noun**, and that **slithy** is an **adjective**?

Twas brillig, and the slithy toves

Did gyre and gimble in the wabe;

All mimsy were the borogoves,

And the mome raths outgrabe.

Lewis Carroll, *Jabberwocky* (1871)



# Exercise: Parts of Speech and Morphemes

| English Example | Morphemes | Parts of Speech |
|-----------------|-----------|-----------------|
| eats            |           |                 |
| waited          |           |                 |
| sleeping        |           |                 |
| eaten           |           |                 |
| Leo's           |           |                 |
| higher          |           |                 |
| smartest        |           |                 |

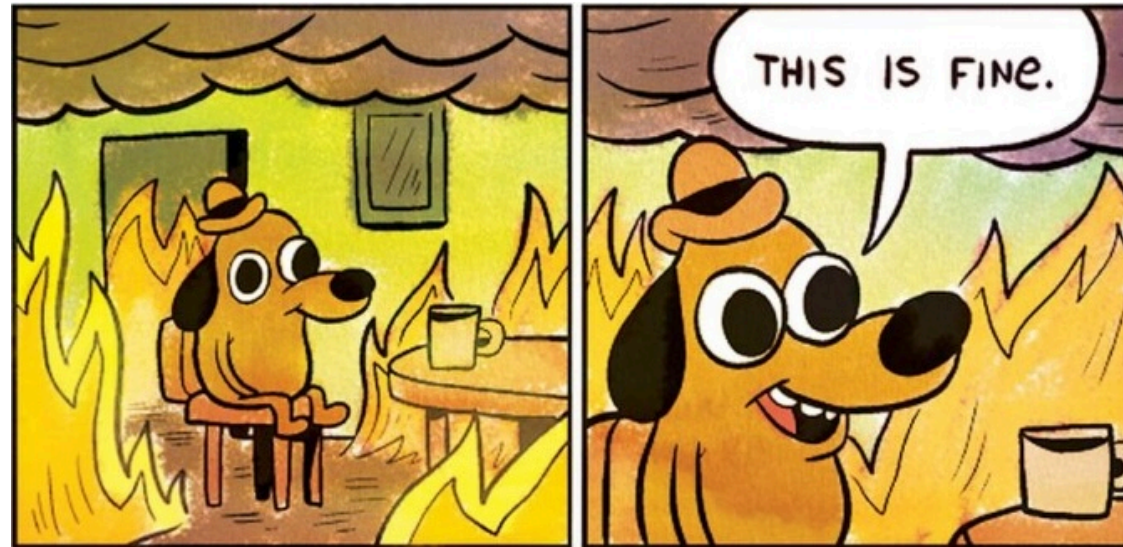
# Exercise: Parts of Speech and Morphemes

| English Example | Morphemes         | Parts of Speech           |
|-----------------|-------------------|---------------------------|
| eats            | <i>eat,-s</i>     | verb, 3rd person singular |
| waited          | <i>wait,-ed</i>   | verb, past tense          |
| sleeping        | <i>sleep,-ing</i> | verb, progressive         |
| eaten           | <i>eat,-en</i>    | verb, perfect             |
| Leo's           | <i>Leo,'s</i>     | (proper) noun, possessive |
| higher          | <i>high,-er</i>   | adjective, comparative    |
| smartest        | <i>smart,-est</i> | adjective, superlative    |

**no homework** is due this week

continue **reading** Sections 4.0, 4.1 and 4.2 of Chapter 4: Morphology from *Language Files*

**readings** Sections 4.3, 4.4 and 4.5 of Chapter 4: Morphology from *Language Files*



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